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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
M.Sc Degree Examination – APRIL 2021

PCH806T – INORGANIC CHEMISTRY – II

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Which one of the following is not a Lanthanide?
   a) Cerium b) Erbium c) Lutetium d) Thorium

2. EDTA is
   a) Monodentate ligand b) Bidentate ligand
   c) Tridentate ligand d) Hexadentate ligand

3. The electronic transitions between orbitals common to both metal and ligand atoms are explained by
   a) IR spectra b) Charge transfer spectra
   c) NMR spectra d) None of the above

4. The word "Nephelauxetic" comes from
   a) Greek b) Latin c) Egypt d) Urdu

5. SEM can achieve resolution better than
   a) 1 nanometer b) 10 nanometer
   c) 100 nanometer d) 1000 nanometer

6. A Bucky Ball is
   a) C60 b) C70 c) C50 d) C40

7. The oxidation state of iron in met-hemoglobin is
   a) 3 b) 2 c) 4 d) 0

8. The correct set of biologically essential elements is
   a) Fe, Mo, Cu, Zn b) Fe, Cu, Co, Ru
   c) Cu, Mn, Zn, Ag d) Fe, Ru, Zn, Mg

9. Nucleons are made up of
   a) Electron b) Proton
   c) Neutron d) Neutron and Proton

10. Which one of the following is a radioactive element?
   a) Uranium b) Aluminum
   c) Silicon d) None of the above

II. Fill up the blanks

11. Molecular orbital theory is a method for determining _____ structure.

12. Orgel diagrams are _____ diagrams.

13. Nanodots are useful in _____

14. Many organisms fix N_2 in nature only _____

15. Nuclear isomerism was first revealed in the reactions on _____

III. Match the following

16. Jahn Teller effect - Manipulation of matter

17. Tanabe-sugano diagram - Spontaneous Symmetry breaking

18. Nano Technology - Radiometric indicator

19. Cytochrome - To predict absortion

20. Radiometric titration - Generation of ATP

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Write a note on Actinides.

22. Give an account on spectrochemical series.

23. Define: Coordination number with example.

24. Write a note on Nephlauxetic effect.

25. Give an account on iron-sulphur proteins.

26. Write a note on Nanotechnology.

27. Briefly explain about quantum wire.

28. Give an account on Heme proteins.

29. Write a note on nuclear isomerism.

30. What do you know about bubble chamber?

31. Give an account on radiometric titration.

32. Give the oxidation states and magnetic behavior of iron in oxy deoxy hemoglobin.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Write a note on Jahn –Teller distortion and explain z-in and z-out cases. **(4)**
    b) Explain Lanthanide contraction. **(3)**

34. a) How orgel diagram is used to predict the spectrum of d^1,d^6,d^4 and d^9 octahedral systems. **(4)**
    b) Write any three notable difference between charge transfer spectra and d-d transition. **(3)**

35. Give the principles of SEM and TEM. **(7)**

36. a) Write a note on carbon ranotubes and its applications. **(4)**
    b) Write a short note on biomedical applications of namotechnology. **(3)**

37. a) Explain the structural aspects of hemoglobin. **(4)**
    b) Write a note on various types of ferredoxins. **(3)**

38. a) Howwill you determine the radioactivity by cloudchamber and bubble chamber method. **(2.5+2.5)**
    b) Write a note on cyclotron. **(2)**

39. a) Write a note on radiometric titrations. **(3)**
    b) Write the differences between hemocyanin and hemeerythrin. **(4)**

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